



Computational Approaches to Alzheimer's Disease

EEG-Based Characterization of Neurophysiological Alterations in Mild Cognitive Impairment During Serious Game Interaction Preliminary Results

Authors and Affiliations

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Background //



Mild Cognitive Impairment (MCI) is an early stage of Alzheimer's disease associated with subtle neurophysiological alterations. EEG, particularly alpha rhythm activity, reflects thalamo-cortical dynamics and is sensitive to early changes in brain function [1]. While most studies rely on resting-state recordings, task-based paradigms may better capture cognitive deficits [2]. Serious games provide ecologically valid environments to evaluate attention, visuomotor, and executive functions. However, their use with low-density EEG systems in MCI populations remains underexplored [3].

Objective //

To analyze EEG signals during resting-state and **SmartMe&You** video game interaction to identify neurophysiological differences between MCI and healthy controls.

Methods //

Fifteen participants were included: nine with MCI (71.6 ± 13.2 years) and six healthy controls (40.1 ± 22.8 years). EEG data were recorded during resting-state (eyes closed, 5 min) and during interaction with the SmartMe&You serious game.



Figure 2. SmartMe&You video game logo

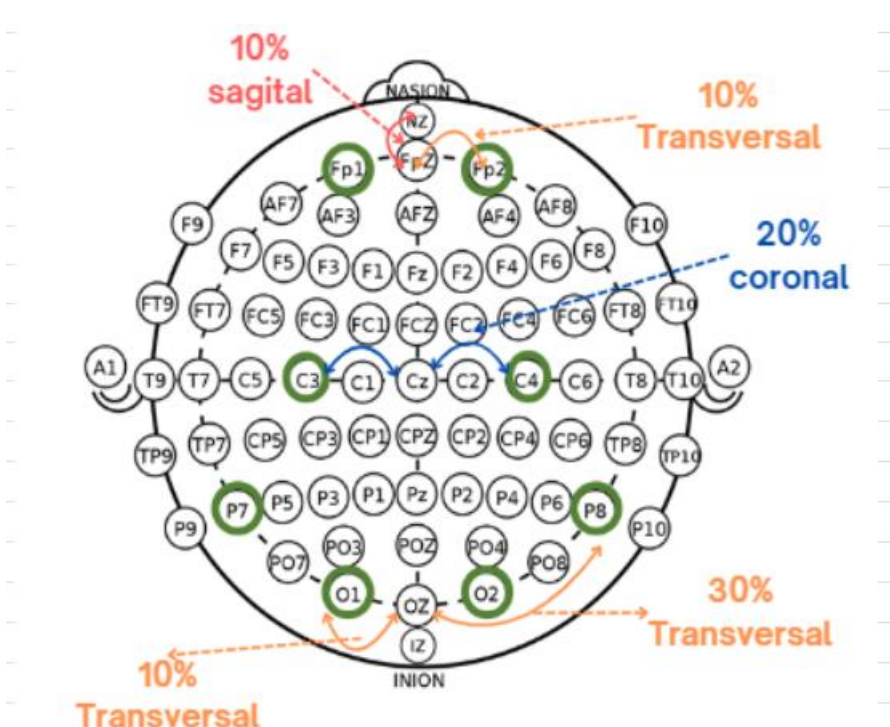


Figure 3. Electrodes used in the study

Signals were acquired using an 8-channel system and analyzed in the alpha band. Behavioral performance was assessed using reaction time and accuracy, and statistical comparisons were performed between groups.

Results //

EEG recordings during the SmartMe&You tasks lasted approximately 12 minutes, and 5 minutes during the closed eyes condition. All signals were processed using the APPLLEE [4] pipeline for artifact removal and preprocessing. Data from 8 channels (Fp1, Fp2, C3, C4, P7, P8, O1, and O2) were segmented into 30-second windows with a 15-second overlap for spectral analysis.

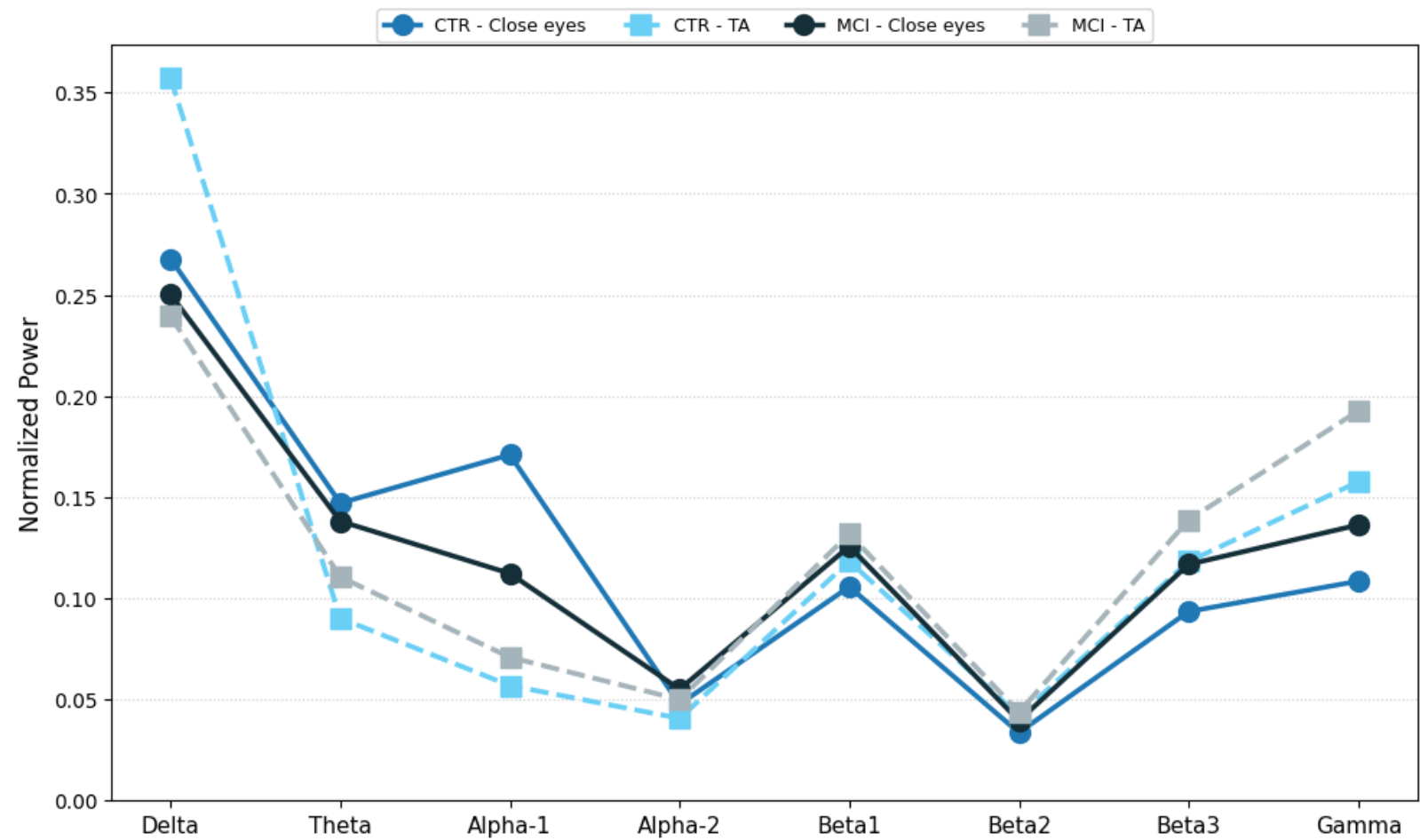


Figure 4. Normalized relative power across canonical frequency bands (Delta to Gamma) for the four experimental conditions: Healthy Controls (CTR - Close Eyes, CTR - TA) and Mild Cognitive Impairment (MCI - Close Eyes, MCI - TA). Data shown as group means (N=15 per group) from one selected channel (e.g., P7). Markers represent mean power for each band; dashed lines indicate SmartMe&You tasks (TA), and solid lines indicate close-eyes rest (Close Eyes).

Also, MCI patients showed an anomalous increase in Beta-3 and Gamma power during tasks in central and occipital regions; this pattern will be further investigated in larger samples.

While Healthy Controls (CTR) showed a significant Delta increase during task performance, MCI patients exhibited a "blunted" response with minimal changes across conditions.

CTR subjects displayed higher baseline Alpha-1 power. Both groups showed task-related Alpha desynchronization, but the magnitude of this reduction was significantly deeper in the CTR group.

Behavioral Results (SmartMe&You Tasks)

Global SmartMe&You video games performance			
	CTR	MCI	p value
Accuracy (%)	87.68 ± 13.77	42.99 ± 34.55	p < 0.001
Reaction time (ms)	403.57 ± 858.02	276.34 ± 269.21	p > 0.05

Table 1. SmartMe&You task metrics: MCI vs. CTR. Mean and standard deviation for behavioral variables. P-values reflect the main effect of diagnosis (MCI vs. CTR) after adjusting for Age through a General Linear Model (sm.stats.anova_lm).

A significant group effect was found for Accuracy during the SmartMe&You tasks. To ensure these differences were not driven by age disparities between groups, an ANCOVA was performed with Age as a covariate. The analysis confirmed that MCI patients performed significantly worse than CTR (p < 0.001) independent of age. In contrast, Reaction Time (RT) showed high inter-subject variability and did not reach statistical significance (p > 0.05), suggesting that the cognitive impairment in this MCI cohort primarily affects task precision rather than processing speed.

Conclusions //

Behaviorally, MCI patients exhibited significantly lower accuracy, a difference that remained robust after controlling for age. Electrophysiologically, healthy controls showed a greater increase in delta power and a stronger suppression of the alpha-1 band compared to the MCI group during the SmartMe&You tasks. Furthermore, the task-related increase in high-frequency activity (Beta-3 and Gamma) observed in both groups warrants further investigation in a larger cohort.

Future Work:

Cohort Expansion | Genetic Analysis | Deep Learning | Explainable AI (XAI)

Acknowledgments //

The authors would like to thank the University of Rome for providing access to the SmartMe&You platform. We also acknowledge the participants for their time and contribution to this preliminary study.

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